

Digitisation of Rendered Images and Texture Sources

1 Purpose and common convention

This note records the image and texture digitisation convention shared by the render and analysis code. The durable render product is a floating-point `.npy` image. An 8-bit TIFF is only a visualisation preview. All measurement analyses form their requested bit depth directly from the stored floating-point image using the same function below; higher-bit TIFFs are not a separate source of truth.

For B bits, define the largest unsigned camera code as

$$M_B = 2^B - 1. \quad (1)$$

For a normalised camera intensity $I \in \mathbb{R}$, the common digitiser is

$$Q_B(I) = \text{round}(M_B \text{ clip}(I, 0, 1)), \quad \text{clip}(z, 0, 1) = \min\{1, \max\{0, z\}\}. \quad (2)$$

The implementation is `numpy rint`: it rounds to the nearest integer, with exact half-way cases rounded to the even integer. The output type is `uint8` for $B \leq 8$ and `uint16` otherwise.

Equation (2) is implemented centrally as `modules.render_outputs.quantise_camera`. It is used for render previews and for the on-the-fly digitised-error analyses in Experiments 1–3.

2 Additive speckle images

Let $c(\mathbf{x})$ be the raw additive coverage field. For a camera pixel P , its pixel-box integration—analytic where available, otherwise SSAA—is

$$\bar{c}_P = \frac{1}{|P|} \int_P c(\mathbf{x}) \, d\mathbf{x}. \quad (3)$$

Disk and Gaussian contributions add in c ; hence overlapping unit-amplitude blobs can give $\bar{c}_P > 1$. This raw coverage is intentionally retained in floating-point additive texture assets and in the float-texture Riley path. No clipping occurs during texture generation or before pixel integration.

Only after the final pixel value has been integrated is coverage mapped to normalised image intensity:

$$s_P = \text{clip}(\bar{c}_P, 0, 1), \quad (4)$$

$$I_P = \text{clip}(I_0 + \gamma(1 - 2s_P), 0, 1). \quad (5)$$

Thus zero coverage is the bright level $I_0 + \gamma$, and saturated coverage is the dark level $I_0 - \gamma$. With the current $I_0 = 0.5$ and $\gamma = 0.4$, these limits are 0.9 and 0.1, respectively. The final camera code is $Q_B(I_P)$ from Equation (2).

The ordering is important:

$$\text{clip}\left(\frac{1}{|P|} \int_P c(\mathbf{x}) \, d\mathbf{x}, 0, 1\right) \neq \frac{1}{|P|} \int_P \text{clip}(c(\mathbf{x}), 0, 1) \, d\mathbf{x} \quad \text{in general.} \quad (6)$$

The code uses the left-hand expression. It is therefore an additive, post-integration saturation model rather than the geometric union of opaque disks.

3 Eggbox images

The eggbox is already an intensity function, rather than a coverage field. In reference coordinates with pitches p_x, p_y , it is

$$I_{\text{egg}}(x, y) = I_0 - \gamma + \frac{\gamma}{2} \left[1 + \cos\left(\frac{2\pi x}{p_x}\right) \right] \left[1 + \cos\left(\frac{2\pi y}{p_y}\right) \right]. \quad (7)$$

The renderer integrates this intensity over the pixel footprint, then applies Q_B only when an integer camera image is required. Eggbox float textures therefore store intensity directly; additive float textures store raw coverage.

4 Float and uint texture sources

For an additive float texture, a texel stores the area-averaged raw coverage $\bar{c}_T$. A uint texture of depth B is derived from that same float asset, not regenerated from the procedural speckle field:

$$T_B^{\text{uint}} = Q_B(I_0 + \gamma [1 - 2 \text{clip}(\bar{c}_T, 0, 1)]). \quad (8)$$

For eggbox textures, whose float value is already intensity,

$$T_B^{\text{uint}} = Q_B(T^{\text{float}}). \quad (9)$$

Consequently, changing the uint source bit depth changes only the quantised texture source supplied to Riley; it does not repeat analytic texture integration. A separate Riley render remains necessary because interpolation of a quantised source is physically a different operation.

5 Persisted outputs and analysis

The standard persisted final-image convention is a normalised floating-point camera intensity I_P after Equation (5) (or the eggbox equivalent). The render helpers save this `float64 .npy` file and, if absent, derive only an 8-bit TIFF preview using Q_8 . For a requested analysis depth B , both candidate and reference float images are independently converted with Q_B before their integer-code difference is calculated. This makes all reported bit-depth metrics reproducible from one canonical floating image, without depending on a TIFF encoder or pre-existing B -bit files.